

Peak-Hour Decision System

Product Requirements Document

for Dark Store Managers

Product Name: Peak Guardian

Team: Thala

Date: February 2026 | Version: 1.0

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Problem Statement

Introduction

During peak hours, Blinkit's high-demand dark stores experience operational chaos. What the customer sees is straightforward: a late delivery, a missing item, or a notification saying "item unavailable" minutes after placing an order. But behind every one of these visible failures is a chain of breakdowns that starts inside the store.

The case study frames this clearly: managers react instead of anticipate, struggle to prioritize issues, lack a single clear control layer, and have limited visibility into peak stress signals. The system functions during average load. It destabilizes under peak pressure.

This PRD is the result of a discovery effort across the entire dark store ecosystem. We started broad, studying riders, pickers, and store managers. We researched how the store operates from inbound receiving to rider dispatch. We examined what breaks first and why. The problem statement below reflects the starting point. The narrowing, and our eventual focus area, comes later as an output of discovery.

Background and Context

Quick commerce in India has grown rapidly. Customers in metro cities now expect groceries and essentials within 10 to 15 minutes. The industry landscape as of early 2026:

- India's quick commerce market is valued at roughly \$3.65 billion (2026), growing at 40%+ annually.
- Blinkit operates approximately 2,027 dark stores (as of Q3 FY26), targeting 3,000 by March 2027.
- Swiggy Instamart operates 600+ dark stores across 100+ cities. Zepto operates 700 to 750.
- Blinkit holds 45% market share, followed by Swiggy Instamart at 27% and Zepto at 21%.
- A high-demand dark store processes 1,200 to 2,000 orders per day across 3,000 to 5,000 SKUs in 2,000 to 4,000 sq ft.
- Evening peak (7 to 10 PM) generates 40 to 50% of all daily orders.

Each dark store runs with 15 to 25 in-store staff and 20 to 60 delivery riders during peak. The full operation, from order receipt to rider dispatch, must complete within 10 minutes. The current technology stack includes a proprietary OMS, a WMS for inventory, and a Dispatch Management System for rider assignment. But managers often supplement these with

WhatsApp groups, Excel sheets, and whiteboards because no single unified interface exists for real-time operational signals.

Why Is the Problem Important

Revenue impact: An estimated 15 to 25% increase in per-order cost during poorly managed peak hours. With a baseline of Rs 60 to 80 per order, that is Rs 15 to 30 extra per order during peak.

Customer impact: Evening out-of-stock (OOS) rates reach 78% across categories. Beverages hit 100% OOS in afternoons, dairy reaches 80 to 83%, fresh produce hits 100% in evenings, snacks reach 88% in afternoons.

Operational impact: When a picker hits an OOS item, they lose 1 to 2 minutes searching for alternatives. At one Blinkit dark store, approximately 60 delivery personnel were observed waiting for orders during peak. Every minute of picker delay multiplies into rider idle time.

Staff impact: Store managers earn roughly Rs 45,000/month. "No work-life balance" is the number one complaint on Glassdoor. Annual turnover runs 30 to 40%.

Scale: At even a conservative 12 phantom inventory incidents per store per shift, that is over 24,000 incidents daily across Blinkit's 2,027 stores. Average inventory wastage runs 2 to 4% of inventory value monthly.

Key Challenges

Across the dark store ecosystem, several structural challenges make peak hours particularly fragile:

- The 10-minute promise leaves zero slack. Each stage (picking, packing, staging, dispatch, delivery) has a narrow time budget. If any one stage slips, the entire promise breaks.
- Order volume surges 3 to 6x during peak. Stores go from 0.5 to 2 orders per minute to 5 to 12 orders per minute in the 7 to 10 PM window.
- Inventory data is scattered. The WMS shows stock levels, but the manager also relies on the OMS, WhatsApp groups, Excel sheets, and physical whiteboards. No single source of truth.
- Store performance degrades when daily orders exceed 1,100 to 1,400 or peak rate crosses 8 to 12 orders per minute.
- External demand amplifiers (IPL matches, rain, festivals, salary week) can spike demand 30 to 70% with little advance warning.

- High turnover (30 to 40% annually) means peak-hour shifts are often staffed by relatively new hires with limited experience.

Scope

The case study is explicit about what we are and are not doing:

We are solving for manager decision clarity during peak hours.

We are not redesigning the supply chain, solving demand forecasting, or changing the delivery model.

In scope:

- Real-time SKU availability tracking (depleting on pick scan, incrementing on restock scan)
- RAG (Red/Amber/Green) status per SKU with depletion velocity
- Auto-OOS logic when stock depletes to zero or "not found" reports cross a threshold
- Near-expiry product visibility at SKU level
- Shelf audit capability to reconcile physical count vs system count
- Peak readiness score with external signal integration

Out of scope:

- Demand forecasting or supply chain optimization
- Rider management or dispatch optimization
- Multi-store dashboards or city-level analytics
- Order-level tracking or customer-facing changes
- Workforce management or shift scheduling

Discovery

Secondary Research

Scale of the Problem in India

- India's quick commerce market is valued at roughly \$3.65 billion in 2026 (Mordor Intelligence), expected to grow at 40%+ annually until 2030 (Bain & Company).
- Consumers spent Rs 64,000 crore on quick commerce in FY25, more than doubling FY24 levels (IBEF).
- Quick commerce now accounts for over two-thirds of all e-grocery orders and 10% of total e-retail spending in India.
- 2.4 million blue-collar jobs anticipated by 2027 in the quick commerce sector.
- Dark store space demand projected to increase from 24 million sq ft (2023) to 37.6 million sq ft by 2027.

Operational benchmarks:

- A mid-sized dark store processes 400 to 1,200 orders/day; top stores exceed 1,800.
- Average order contains roughly 6 SKUs; pick target is under 2 minutes per order.
- Pickers operate under 15 to 20 second per-item speed targets.
- SKU counts range from 2,500 at lean stores to 6,000+ at Blinkit, up to 17,000+ at megapod formats.

The OOS problem at scale:

Category	Evening OOS Rate
Fresh Produce	100%
Snacks	88%
Dairy	80-83%
Beverages	82%
Cold Brew Coffee	82%
Energy Drinks	79%
Oat Milk	77%

- 60 to 70% of cart items are repeat purchases, making OOS particularly damaging to loyalty.
- Inventory discrepancy (system shows stock, physical stock is zero) is the number one failure mode.
- Roughly 70% of peak-hour failures trace back to pre-existing internal inefficiencies that get exposed by demand spikes.

Existing Solutions and Initiatives

What exists today:

- WMS-based inventory tracking with barcode scanning, syncing with the customer app reportedly every minute.
- Auto-delisting when stock hits zero.
- AI-driven demand forecasting at SKU-store level (up to 90% accuracy at neighbourhood level).
- Automated replenishment triggers when stock drops below thresholds.
- Pick-path optimization on handheld devices.
- Three-checkpoint verification: scan at pick, scan at pack/billing, scan at rider handoff.

Where these fall short:

- The WMS shows current stock levels, not depletion velocity or time-to-zero.
- No RAG status system to flag SKUs trending toward stockout before they hit zero.
- No mechanism to catch phantom inventory until a picker reports "not found".
- Shelf audit happens during low-traffic periods, never during peak prep.
- No single screen combining inventory status, "not found" frequency, and near-expiry alerts.
- Managers still rely on a patchwork of OMS, WhatsApp, Excel, and whiteboards.

Opportunities for Improvement

1. Existing scan events are underused. Every pick and putaway already generates a scan event. These events are consumed by the WMS for stock count updates but are not aggregated into a real-time operational view for the manager.
2. Depletion velocity is computable but not surfaced. The data to calculate how fast a SKU is being consumed (picks per hour) already exists in the scan event stream. No system currently computes or displays this for the manager.

3. "Not found" reports contain a signal that is currently ignored. Pickers report items as "not found" during order picking. These reports sit in exception logs. Nobody aggregates them in real time to detect phantom inventory patterns.
4. Pre-peak preparation has no structured process. Managers do informal walkabouts before peak. There is no system-guided audit workflow for high-velocity SKUs, and no score or checklist to assess readiness.
5. Near-expiry data exists but is not linked to peak-hour decision-making. Batch and lot expiry dates are recorded during the GRN process. This data is not surfaced to the manager alongside availability status.
6. External demand signals are available but not integrated. IPL schedules, weather forecasts, and festival calendars are publicly available data. No current system feeds these into the manager's peak preparation.

Challenges to Address

1. Data latency: scan events must reflect on the dashboard within seconds, or the manager's view is stale.
2. Adoption under stress: any new tool must be glanceable (under 5 seconds) and require minimal interaction.
3. Physical count accuracy: shelf audits are manual and error-prone. The system should highlight mismatches, not blindly trust.
4. Phantom detection threshold: too low causes false positives, too high means too many failed orders before action.
5. Store layout variability: each store has a unique shelf layout. The dashboard must adapt to store-specific configurations.

Primary Research

- We researched three stakeholder groups inside the dark store to understand where peak-hour failures originate and where intervention has the highest leverage.
- Sources: Delivery partner interview (Blinkit, Bangalore), Play Store and Glassdoor reviews, store manager insights from Glassdoor/Indeed, YouTube and Reddit research, field visits.
- **A. Rider Perspective**
 - A Blinkit rider in Bangalore (split shifts, 6 AM to 1 PM and 5 PM to 10 PM) described the following: Order packing is too slow during peak, with riders waiting 30+ minutes. Dispatch is manual (picker calls out rider's name). On busy days, roughly 60 delivery personnel were observed waiting at a single store. No formal penalty for late delivery, but if a customer reports a missing item, the rider loses the full product value from earnings (example: Rs 500 on Rs 5,000 daily earning). Most riders skip delivery photos due to time pressure, losing disputes by default when they arise. The rider-manager relationship is transactional; the rider only talks to the manager when there is a problem. When asked about interviewing the manager, the rider suggested coming "after 11 PM," confirming the manager is occupied through the entire day.
 - Key limitation: The rider's view is narrow. "Scan in, wait, get called, deliver, come back." No visibility into store-side operations.
- **B. Picker Perspective (Play Store reviews, Glassdoor)**
 - Items physically placed in wrong aisles, making it impossible for new pickers to meet the 3-minute First Pick window. HHT (handheld device) shortage means available pickers cannot work during peak. Over 70% of QR codes on shelves are faded and unscannable at poorly managed stores. One hour of training is insufficient; new pickers are expected to match veteran speed immediately. Pickers are financially penalized for "missing items" that are inventory errors. When a picker hits an OOS item, the workflow breaks: search, wait for substitution approval, or flag the issue. Aisle congestion from multiple pickers converging on the same high-demand shelves drops productivity by up to 25%.
- **C. Store Manager Perspective (Glassdoor, Indeed, field research)**
 - On overload: Managers personally pick and pack when a picker is absent during peak. Time splits during evening peak: people management 30 to 40%, inventory 25 to 30%, order monitoring 20%, admin 10 to 15%. They handle picker assignment, rider queries, substitutions, refunds, escalations, and queue monitoring simultaneously, with no

priority signals. Confirmed: "Early mornings and dinner time are the busiest, I don't even get a chance to check my phone."

- On inventory blindness: No way to see which SKUs are trending toward zero before they hit zero. Cannot distinguish real stockout from phantom inventory until multiple pickers report the same item. Near-expiry products slip through. Data lives across 5+ disconnected systems.
- On the decision gap: Experienced managers proactively pause orders when backlog hits 1.5x capacity. Inexperienced managers miss this entirely. No data-backed threshold exists for when to throttle.

[GAP] *We do not yet have a direct, sit-down interview with an active dark store manager describing their workflow in their own words. The manager perspective is reconstructed from Glassdoor reviews, job postings, and indirect observations. A dedicated manager interview would strengthen this section.*

Competitor Analysis and Limitations

Common limitation across all platforms: Every player treats inventory as a number (current stock count). None treats it as a signal stream (velocity, trajectory, phantom risk, readiness). No platform has a pre-peak readiness workflow or external amplifier integration for the store manager.

Platform	Dark Stores	Key Approach	Limitation
Blinkit	~2,027 (Q3 FY26)	WMS + barcode scanning, AI demand forecasting, FIFO/FEFO, auto-delisting at zero stock	No predictive RAG status. Manager discovers OOS only on picker "not found" report. No unified real-time dashboard.
Zepto	700-750	76-second pack time claim. Software re-optimizes layouts. Strict rider access control.	No documented real-time manager dashboard for SKU depletion signals. Same phantom inventory problem.
Swiggy Instamart	600+	Uses Swiggy's delivery network. Integrated food + grocery.	Less focused on dark store operational tooling. Smaller store count.
Amazon Tez / Flipkart Minutes	Emerging	Backed by massive logistics infrastructure.	Still early stage; significantly smaller dark store networks.

Narrowing the Problem: From Ecosystem to Store Manager

After researching all three stakeholder groups, we needed to decide where to focus. The case study's visible symptoms (delivery delays, wrong items, rider clustering, rising refunds) could be attacked from multiple angles. We deliberately explored each before converging.

Why not solve for the rider?

Consideration	What We Found
Riders are store-tethered	Blinkit riders must physically scan in at the store. They cannot float between stores to rebalance demand.
Rider's view is narrow	"Scan in, wait, get called, deliver, come back." They do not see what happens inside the store.
Rider idle time is a symptom, not a cause	60 riders waiting at one store means orders are not being packed fast enough, which means picking is failing, which means inventory is wrong.
Solving for rider wait improves rider experience	But does not prevent the upstream failure that caused the wait.

Why not solve for the picker?

Consideration	What We Found
Picker sees only one order at a time	No visibility into the next 2 hours of demand.
Picker can only execute or report problems	Cannot throttle categories, push substitutes, or request emergency refills.
Fixing picker experience improves single orders	Does not fix system-level risk across hundreds of orders.
Picker cannot anticipate demand shifts	Cannot predict an IPL-driven snack surge or hide categories from the app.

Why the store manager is the right user:

Consideration	What We Found
Impact at scale	A single throttling decision affects 50+ orders in the next hour.
Authority for strategic intervention	Only the manager can throttle a category, push substitutes, prioritize shelf verification, or request emergency refills.

Human integration layer	The manager's brain is currently the glue between OMS, WMS, WhatsApp, Excel, and whiteboards. They are the de facto control layer.
The 25-30 min gap	Between the first picker failure and manager action is exactly the gap we close. This gap causes 15 to 50 failed orders per night.
Peak readiness is a manager-level concept	Pickers and riders cannot assess store-wide readiness.

We also ran the research through a structured prioritization exercise, asking: What is the single most repeated issue? Where does it happen? When does it become visible? Who is affected first? What decision becomes difficult? What information is missing?

Every path led to the same convergence point: **inventory mismatch at the SKU level, invisible to the manager until a picker hits it mid-order during peak.**

Refined Problem Statement

During peak hours, dark stores frequently show "in-stock" SKUs that are physically missing or misplaced, breaking the pick flow mid-order and triggering cascading SLA failures.

Store managers lack predictive visibility into SKU-level depletion, forcing them to discover fulfillment failures only after a customer's delivery window has already begun.

This reactive environment turns every evening peak into operational chaos, where the 78% OOS rate becomes visible one broken order at a time, with no system-level signal to throttle demand or redirect resources before the problem compounds.

Hypothesis

If the store manager has a real-time, scan-based dashboard showing live SKU availability with predictive depletion signals (RAG status - Red/Amber/Green), automatic OOS detection, and a pre-peak readiness score, they will be able to:

- Make proactive restocking decisions before peak hours
- Detect phantom inventory before pickers encounter it
- Reduce pick-failure-driven SLA breaches by at least 40%

Assumptions(for the solution design)

1. The dark store has the ability to manually control SKU listing/delisting on the customer-facing app.
2. The current dashboard does not show live RAG status or predictive depletion signals at the SKU level.
3. As a PM, we have access to the dark store's inventory data and scan event streams.
4. Pickers already scan items during picking. We can use existing scan events.
5. Restocking staff already scan items during putaway. We can use these scan events too.
6. The delivery partner network is not the primary bottleneck during peak hours (approximately 60 riders wait at a single store during peak).
7. Our primary focus is daily peak hours (7 to 10 PM) in metro-city Blinkit stores. Event-driven amplifiers (IPL, rain, festivals) are accounted for as signals within the readiness score, but we are not designing for extreme once-a-year scenarios like Diwali warehouse overload or large-scale gig worker strikes.

What the Manager Needs

The manager must answer one key question before peak hours:

"Will my current inventory survive tonight's demand?"

To answer this, they need three things:

A. Visibility: What is the real state of my inventory right now?

- Not just system counts, but which counts can be trusted.
- Which SKUs have been reported "not found" today.
- Which items are approaching expiry.
- When each zone was last physically verified.

B. Predictability: What is about to go wrong in the next 3 hours?

- Which SKUs are depleting faster than the restock rate.
- How long until each high-velocity SKU hits zero at the current rate.
- Are there external demand amplifiers tonight (IPL, rain, festival, weekend).
- Is today's order volume tracking ahead or behind same-day-last-week.

C. Actionability: What should I do about it?

- Priority-ranked list of problems (not all data shown equally).
- Clear recommended action per problem (verify shelf, restock, delist, throttle).
- One-tap execution where possible.
- Tradeoff clarity for each decision.

User Persona and Pain Points

Rajesh, Dark Store Manager, Blinkit (Bangalore)

Attribute	Detail
Age	26
Education	Graduate (non-technical)
Previous experience	2 years retail operations
Salary	Rs 45,000/month (Rs 5.35L/year)

Work hours	6:30 AM to 11:00 PM, 6 days/week
Store	3,000 sq ft, ~6,000 SKUs, 5 pickers per shift
Daily orders	600-800 (normal), 1,000+ (festivals)
Devices	Personal smartphone + shared tablet (WMS) + HHT scanner
Reports to	Area Manager (manages 5-8 stores remotely)
Staff under him	15-25 in-store + 20-60 riders during peak
Top complaint	"No work-life balance" (Glassdoor)
Only free	After 11 PM (confirmed by delivery partner)

What Rajesh cannot do today:

- See which SKUs will run out during tonight's peak.
- Distinguish phantom inventory from real stockout until 3+ pick failures.
- Know if today's demand is tracking above historical patterns.
- Know if IPL or rain is expected tonight and how it affects specific categories.
- Run a systematic pre-peak audit of high-velocity zones.
- See all problems in one place, ranked by urgency.

Current User Journey

The 10-minute delivery timeline: Order placed > Picking (2 to 6 min) > Packing (1 to 3 min) > Rider pickup (0 to 2 min) > Delivery (4 to 8 min). Zero slack.

6:00 PM — Manager reviews yesterday's KPIs, does a quick walkthrough. No systematic way to verify if system counts match physical shelves. No signal about what is coming tonight.

7:00 PM — Peak begins. Order rate hits 5 to 12 per minute. All pickers active. Manager in combat mode.

7:15 PM — First "not found" report. Picker goes to shelf for Amul Butter, system shows 8 units, shelf is empty. Report buried in WMS exception log. Manager does not see it.

7:20 PM — Two more pickers report the same item missing. Three customer orders delayed. Item still shows "available" on the app.

7:45 PM — Manager notices through WhatsApp or a physical walkabout. Manually checks shelf, confirms system count is wrong, triggers delisting (multi-step process).

8:30 PM onward — Same pattern repeats for 5 to 10 high-velocity items. Each causes 3 to 5 delayed orders before being caught.

10:00 PM — Peak ends. Manager discovers 12+ phantom incidents during reconciliation. Too late to fix.

The core gap: 25 to 30 minutes between first "not found" and manager action. 5 to 8 failed orders per phantom SKU. Multiply by 5 to 10 phantom SKUs per peak, and the damage is 25 to 80 failed orders per night.

Solution

Vision Statement

A single-screen, real-time dashboard that gives the store manager predictive visibility into every SKU's availability status, so they can act before stock runs out, not after customers are already waiting.

Powered by existing scan events (pick scans deplete, restock scans increment), it shows RAG status per SKU updating in real time. The manager sees problems forming: amber SKUs trending toward red, "not found" reports clustering on a specific item. And they can act: trigger a restock, mark as OOS, or verify the shelf.

No new hardware. No change to picker workflow. A software layer that turns existing scan data into actionable signals. The system shifts the manager from reactive firefighting to proactive prevention.

Design Principles

The case study specifies that peak-hour design must work under extreme stress. Our design follows four principles:

1. **Glanceable.** Manager must understand status in under 5 seconds. No dense tables, no long reads. Colour codes (RAG) and counts do the talking.
2. **Action-oriented.** Every screen leads to a decision, not just data. Problems surface with recommended actions attached.
3. **Low cognitive load.** Problems surface automatically. The default view is filtered to Amber and Red only. The manager does not hunt through menus.
4. **Stress-tested.** Works when the manager is handling 8 to 12 orders per minute, pickers are calling out, and riders are waiting. Minimal taps, large touch targets, clear hierarchy.

Product Structure

The dashboard is organized into three tabs, each serving a distinct function:

Tab 1: Live Availability (default view)

- KPI cards at top: count of Red/OOS SKUs, Amber/At-Risk SKUs, Near-Expiry units.
- SKU table in the middle: RAG status dot, system count, depletion velocity (picks/hr), time-to-zero, not-found report count, near-expiry units, action buttons (Restock / Mark OOS / Verify).

- Default filter: Amber and Red only (togglable).
- Activity feed at bottom: scrolling timeline of events ranked by priority.

Tab 2: Shelf Audit

- Zone selector.
- SKU list for selected zone with system count, physical count (manual input), match/mismatch status, and Sync button.
- Progress bar: X/Y audited, Z mismatches.

Tab 3: Peak Readiness (surfaced at 5 to 6 PM)

- Score: 0 to 100 with status label (Ready / At Risk / Vulnerable / Critical).
- Component breakdown (6 factors).
- External amplifier badges (IPL, rain, festival, weekend).
- Top 3 recommended actions.

Problem-Solution Mapping

Problem (from research)	Solution Feature	How It Works
Manager discovers OOS only after picker fails (25-30 min gap)	Real-time RAG Status	Amber warnings at 3-10 units. Gives 15-30 minutes to act before Red.
Phantom inventory: system says in-stock, shelf is empty	Auto-OOS on "not found" pattern	3 "not found" reports in 1 hour triggers auto-OOS and auto-delist.
No way to see depletion velocity or time-to-zero	Depletion Velocity Tracking	Tracks picks/hr per SKU. Projects: "Amul Milk, 8 units, 3/hr, zero by 8:40 PM."
No single screen for inventory health	Live Availability Tab	One table: RAG status, count, not-found, near-expiry, last scan. Default: Amber & Red.
No way to verify system count vs physical	Shelf Audit Tab	Select zone, type physical counts, system highlights mismatches, one-click sync.
Near-expiry slips through during rush	Pre-Peak Expiry Audit	6 PM: surfaces SKUs expiring within 24-48 hours, filtered to high-velocity items.
No predictive signal about tonight	Peak Readiness Score	0-100 score: inventory health, depletion, phantom risk, external amplifiers.
Decision overload, no prioritization	Priority-Ranked Activity Feed	Events ranked by urgency. Red floats to top. Shows 2-3 decisions needing attention.

Key Features

Feature 1: RAG Status with Depletion Velocity

Status	Condition	Action Signal
GREEN	> 10 units, stable depletion	No action needed
AMBER	3-10 units, OR depleting fast	Needs attention: restock or verify
RED	0-2 units, OR phantom (not-found x3)	Immediate action: delist or verify

Depletion velocity is what makes RAG predictive rather than reactive. A SKU at 12 units but depleting at 6/hour shows as Amber because it will be Red in 2 hours. A SKU at 4 units but depleting at 0.5/hour stays Green because it will last the entire peak.

Data source: WMS pick scan events (already generated on every pick). Rolling 30-minute window for velocity calculation. Updates on every scan event.

Feature 2: Phantom Inventory Detection

Signal	Action
"Not found" x3 in 1 hour	AUTO-OOS triggered, auto-delisted from customer app
"Not found" x2 in 2 hours	MONITORING: dashboard flags as suspect
System count > 5 but last successful pick > 4 hours ago	STALE: "verify shelf" prompt

The threshold of 3 was chosen deliberately. One report could be a picker error. Five means 5+ customers already impacted. Three in one hour balances sensitivity against accuracy. If 3 different pickers cannot find the item, it is almost certainly not there. A "Reinstate" button lets the manager reverse auto-OOS if items are located.

Feature 3: Peak Readiness Score

A single 0 to 100 score generated at 5:00 PM and 6:00 PM before the 7 PM peak.

Component	Weight	What It Measures
Inventory health	30%	% of top-100 SKUs that are Green
Depletion trajectory	20%	How many SKUs will hit zero by 10 PM at current rate
Phantom risk	15%	Count of SKUs with "not found" or "stale" flags

External amplifiers	15%	IPL, rain, festival, weekend, salary week
Replenishment status	10%	Is today's restock truck delivered and shelved?
Audit freshness	10%	When was the last shelf audit of high-velocity zones?

Score bands: 80 to 100 is Ready. 60 to 79 is At Risk. 40 to 59 is Vulnerable. Below 40 is Critical. The score surfaces the top 3 recommended actions so the manager knows exactly where to focus before peak.

Feature 4: Pre-Peak Expiry Audit

At 6:00 PM, the system generates a checklist of SKUs with units expiring within 24 to 48 hours, filtered to high-velocity items. The manager physically checks shelves, pulls near-expiry products, and marks them as removed or discounted. Data source: batch/lot expiry data already in WMS from the GRN process.

Feature 5: Unified Activity Feed

A single chronological notification stream replacing the fragmentation of WMS exception logs, WhatsApp, and verbal updates. Four priority levels:

- CRITICAL: Not Found x3 triggers auto-OOS. Vibration + persistent banner.
- WARNING: SKU hits Amber, or Not Found x2. Single vibration.
- INFO: Restock scan completed, SKU goes Green. Silent.
- ACTION: Pre-peak audit reminder, expiry audit due. Badge only.

Feature 6: Shelf Audit Mode

Manager selects a zone, walks to shelves, types physical counts. System compares with WMS and highlights mismatches. Mismatch is a flag, not an auto-correction. Manager clicks "Sync" to confirm, preventing accidental overrides. RAG status recalculates immediately.

Feature 7: Manager Action Options

Action	What It Does	When to Use
Verify Shelf	Assigns picker to physically count at that location	System count seems wrong or "stale" flag
Restock	Triggers restocking task for that SKU	Count is low but physical stock exists in backroom

Mark OOS	Immediately delists SKU from customer app	Shelf confirmed empty
Throttle Category	Temporarily hides category from customer app	Multiple SKUs in a category are at risk
Push Substitutes	Surfaces alternatives to customers automatically	High-demand SKU is OOS but substitutes exist
Request Emergency Refill	Triggers warehouse restock request	Stock critically low, next restock too far
Reinstate	Reverses auto-OOS if items found on shelf	Auto-OOS was a false positive

Feature 8: Picker Confidence Indicator (Phase 2)

On the picker's handheld, alongside the shelf location, show a confidence indicator: High (green dot), Check Shelf (yellow dot), or Likely Missing (red dot). Based on system count, recent "not found" history, time since last successful pick, and whether the zone was recently audited.

Deferred to Phase 2 because it requires picker app UI changes, which is a cross-team dependency. The MVP is 100% manager-facing.

Prototype link

<https://peak-guardian-flow.lovable.app/>

How the Picker Fits In

The picker is not the primary user of this system. **The picker is the execution layer that enables manager decisions to become real-world action.**

Picker responsibilities in this system:

- Receive verification tasks (assigned by manager via dashboard).
- Physically count inventory at the assigned shelf.
- Update actual quantity.
- Close the data gap between system and reality.

During active order picking (Phase 2):

- Picker sees confidence warnings on handheld.
- Pre-loaded substitute suggestions if the SKU is flagged.
- If the SKU is missing: one-tap substitute, customer notified instantly, order flow continues.

Role clarity:

Manager	Picker
Strategic decision maker	Executes decisions
Prevents failure at scale	Feeds ground-truth data
Controls throttling and prioritization	Ensures operational continuity
Sees the whole store	Sees one order at a time

Results for End-User**For the store manager:**

- Sees problems forming before they impact customers.
- One screen replaces OMS + WhatsApp + walkabouts.
- Can run a focused shelf audit of high-velocity zones before peak.
- Near-expiry visibility prevents FSSAI compliance violations.
- Peak readiness score gives a 5-second answer at 6 PM.

For pickers:

- Fewer "not found" encounters because phantom inventory is caught earlier.
- Smoother pick flow during peak.
- Phase 2: confidence indicator warns before walking to an empty shelf.

For riders:

- Orders packed faster because pick failures are reduced.
- Less idle waiting time.
- Fewer customer complaints means fewer earnings deductions.

For customers:

- Fewer orders affected by OOS mid-pick.
- More accurate availability on the app.
- Reduced delivery delays during peak.

The Bold Idea: Assisted One-Tap Actions

This section encapsulates what the ideal product could be like. The suggested solution takes this for direction setting, for what could come out as a solution at various layers and phases, to provide true utility of the solution.

The system does not just show problems. It pre-computes the recommended response and presents it as a one-tap decision.

The backend monitors live signals every 60 seconds: how fast orders are being picked, how long riders are waiting, which SKUs are running low, whether packing is becoming a bottleneck. It computes a stress score and decides what the manager needs to see urgently.

The manager sees one clean screen in three sections:

- Top: The 3 most critical issues right now, ranked by impact on delivery time. Each has a suggested fix as a button. Example: "12 riders waiting, 9 orders unpacked. Move 2 pickers to packing?" One tap, done.
- Middle: A 20-minute ahead prediction. If a spike is coming based on order rate or external signals (rain, IPL), the manager sees it before it hits.
- Bottom: Floor-level view of which picking zones are congested or understaffed.

Why this is bold: The manager does not lose authority. They skip the step of figuring out what to do, which is where most time is lost during peak. The system pre-computes; the manager approves or overrides.

Additionally, the system tracks peak-hour pick frequency by SKU and gives the manager a weekly recommendation to move high-demand items to easier-access locations before the week's peak evenings.

Implementation Plan

Phase	Scope	Stores
Phase 0: Data Validation	Verify scan event stream latency, "not found" log format, expiry data availability.	3-5 (Bangalore, Pune)
Phase 1: MVP	Live Availability tab: RAG status, depletion velocity, auto-OOS, activity feed.	5 high-volume stores
Phase 2: Readiness	Shelf Audit tab, near-expiry column, KPI cards, Peak Readiness Score, external amplifiers.	50 stores
Phase 3: Intelligence	Picker Confidence Indicator, one-tap assisted actions, historical trend analysis.	200 stores
Phase 4: Scale	Network-wide rollout. Cross-store benchmarking. Area manager view.	500 to 2,000 stores

MVP includes: Live Availability tab (RAG status, depletion velocity, time-to-zero), Auto-OOS on "not found" x3, Activity feed (priority-ranked), Basic action buttons (Restock, Mark OOS, Verify Shelf).

MVP does NOT include: Shelf Audit tab, Peak Readiness Score, External amplifiers, Picker Confidence Indicator, One-tap assisted actions, Cross-store benchmarking.

MVP success criteria: Manager daily active usage > 90% in pilot stores within 4 weeks of rollout.

Trade-offs and Limitations

Trade-off	What We Chose	What We Gave Up	Why
Breadth vs depth	Inventory visibility only	Not solving rider dispatch, scheduling, or routing	Inventory mismatch is the #1 failure mode. Highest impact first.
Automated vs manual	Auto-OOS after 3 "not found"	Manager does not approve every delist	Speed matters during peak. Reinstate button is the safety valve.
Predictive vs reactive	Depletion velocity projects future state	Will not catch unprecedented spikes (first monsoon rain, viral trends)	Covers 90%+ of predictable scenarios. Unprecedented events are out of scope for V1.
Manager-first vs picker-first	Dashboard for manager in Phase 1	Picker confidence indicator delayed to Phase 2-3	Manager has system-level authority. Picker changes need cross-team work.
Simple score vs complex model	Weighted formula for readiness	Not using ML in V1	Weighted formula is transparent and ships in weeks. ML can layer later.
Top SKUs vs all SKUs	RAG for high-velocity SKUs first	Long-tail SKUs not covered initially	Top 100 drive 60-70% of orders. Covering all 6,000 adds compute without proportional impact.

Constraints from the case study:

- Cannot reduce growth demand.
- Cannot drastically reduce SKU count.
- Cannot expand physical store size.
- Must work within realistic operational limits.
- Must be scalable across multiple stores.

Challenges and Mitigation

Challenge	Mitigation
Data latency from scan events	Target sub-5-second latency using existing real-time infrastructure (Amazon SQS). Staleness indicator if data > 30 seconds old.
Manager adoption under stress	Default filter: Amber & Red only. Problems surface automatically. Launch as pre-peak tool (5-6 PM) first to prove value.

False positives from auto-OOS	Threshold of 3 in 1 hour. Reinstate button if wrong.
Store layout variability	Zone/shelf mapping from existing WMS bin location data. No manual config needed.
Cross-team dependency for picker features	Picker features deferred to Phase 3. MVP is 100% manager-facing.
External API reliability	Amplifiers are additive, not core. Static festival calendar as fallback. Dashboard works fully without external signals.
Store connectivity issues	Local cache of last-known state. Offline audit checklist. Staleness indicator shown to manager.

Success Metrics

North Star Metric: Evening peak fill rate (% of orders during 7 to 10 PM where all items were available and fulfilled without substitution). Baseline from research: roughly 22%.

Primary Metrics:

Metric	Current Baseline	Target (3 months)	Target (6 months)
Phantom incidents per shift	~12 per store	~7 (40% reduction)	~5 (60% reduction)
Time to detect OOS	~15 minutes	< 5 minutes	< 2 minutes
Pick success rate (peak)	~82%	88%	93%+
Pre-peak audit completion	0% (no process)	50% of stores	>80% of high-velocity zones before 7 PM
Manager daily active usage	0% (tool doesn't exist)	>70% pilot stores	>90% network-wide

Secondary Metrics:

- SLA breach rate (7 to 10 PM).
- Rider idle time at store during peak.
- Refund/cancellation rate during peak.
- Peak Readiness Score accuracy (correlation with actual fill rate; target R-squared > 0.5 at 3 months).
- "Not found" reports per shift.

A/B Test Design:

- Control: 50 stores, no dashboard (current workflow).
- Treatment: 50 stores, Peak Guardian deployed.
- Duration: 8 weeks minimum.
- Comparison: evening fill rate, phantom incidents, OOS detection time, failed orders.
- Significance: $p < 0.05$.
- Selection: matched by city, order volume, and store maturity.

Userflow

Before and After: Manager Workflow

Time	Before (Current)	After (With Peak Guardian)
5:00 PM	No preparation.	Peak Readiness Score computed. Sees IPL tonight, dairy zone unaudited. Acts on top 3 recommendations.
6:00 PM	Quick walkthrough, gut feel. No data on what is coming.	Expiry audit checklist auto-generated. Shelf audit of Zone B (15 min).
7:15 PM	First "not found" from picker. Buried in WMS exception log.	Dashboard shows "not found" instantly. Manager sees it in real time.
7:45 PM	Manager notices via WhatsApp. Walks to shelf. Multi-step delisting.	Not Found x3 at 7:29 triggers AUTO-OOS. Zero manual intervention.
8:30 PM	5-10 phantom SKUs discovered. 15-50 failed orders.	1-2 phantom SKUs caught early. 3-8 failed orders.
10:00 PM	Discovers 12+ incidents post-peak. Too late to fix.	Incidents resolved during peak. Shift summary auto-generated.

Flow 1: Real-time Monitoring During Peak

Manager opens the dashboard at 6:30 PM. KPI cards show the count of Red, Amber, and Near-expiry SKUs. Default view: Amber and Red only.

- Sees an Amber SKU (Maggi at 5 units, depleting 4/hr). Taps "Restock". After restocking, count increases and it goes Green.
- Sees a Red SKU with "Not Found x3" (Amul Butter, phantom). Walks to shelf to verify, or leaves as OOS. If confirmed empty, stays OOS. If found misplaced, taps "Reinstate" and the counter resets.
- Sees a Near-Expiry flag (6 units of Amul Milk expiring in 18h). Pulls those units for markdown before peak.

Flow 2: Pre-Peak Shelf Audit (6:30 to 7:00 PM)

Manager switches to the Shelf Audit tab. Selects Zone B (dairy, highest velocity before peak). Walks to B1/Shelf 1 and physically counts items. Types the count into the dashboard.

- If physical count matches system: green (Match).
- If it does not match: red highlight (Mismatch). Manager taps "Sync" and system count updates to physical truth. Progress bar shows: "6/10 SKUs audited, 3 mismatches synced."

Flow 3: Auto-OOS Trigger During Peak

7:20 PM: Picker 1 reports "not found" for Coca-Cola 750ml. Dashboard: Not Found = 1.

7:25 PM: Picker 2 reports "not found" for same SKU. Dashboard: Not Found = 2 (monitoring).

7:29 PM: Picker 3 reports "not found". Threshold crossed (3 in 1 hour). RAG auto-changes to Red. SKU auto-delisted from customer app. Activity feed shows the event. Manager can tap "Reinstate" if needed.

Result: roughly 14 minutes total (vs 25 to 30 today). Failed orders: 3 (vs 5 to 8 today).

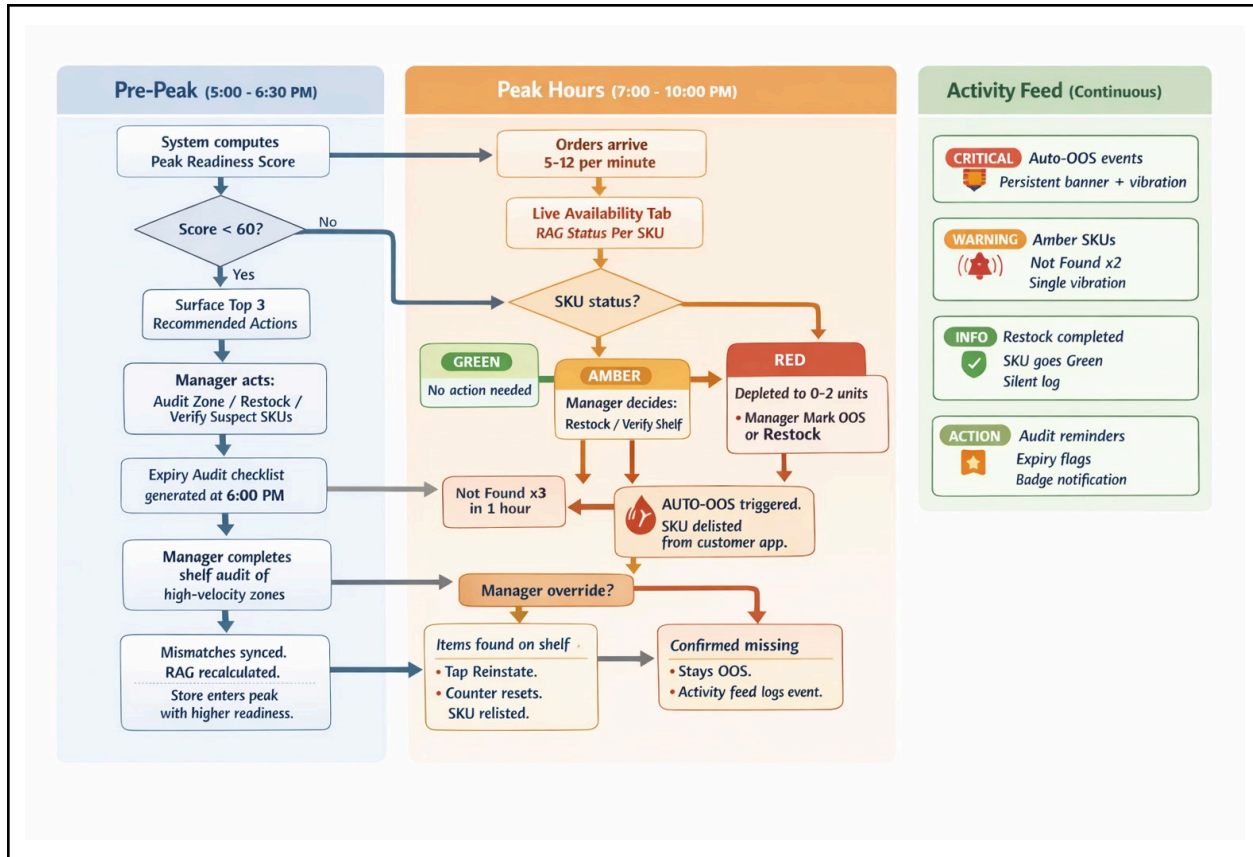
Flow 4: Peak Readiness Check

5:00 PM: System computes Peak Readiness Score. Score: 62/100 (At Risk).

Breakdown: Inventory health 71 (8 Red SKUs), Depletion trajectory 55 (12 SKUs will hit zero by 9 PM), Phantom risk 70 (3 suspect SKUs), External amplifiers 45 (IPL tonight), Restock status 80 (truck arrived, 90% shelved), Audit freshness 40 (dairy not audited today).

Top 3 actions: (1) Audit dairy zone (last audit 18h ago). (2) Restock Lay's Classic (6 units, 4/hr depletion). (3) Verify Toned Milk (not found x2 today). Manager acts on these before 7 PM.
Score recomputed at 6:00 PM: 78/100 (improved).

Flow Diagram



References

Primary research sources:

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- Store manager research and Glassdoor/Indeed reviews analysis
- Blinkit Picker App Play Store reviews analysis
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- California Management Review (CMR), article co-authored by Zepto's Senior Director of Operations
- Savills India report (Dec 2025), dark store space demand
- Blinkit engineering blog: barcode scanning, inventory system architecture
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- JM Financial estimates: picking time benchmarks
- FSSAI regulations: shelf-life compliance mandates
- PissedConsumer: Blinkit customer reviews and ratings
- Glassdoor: Blinkit/Zepto employee reviews and salary data